

AURÉLIEN PASSELAIGUE

Full-Stack / R&D Engineer Intern (Final-year) | Offline-first PWA, IoT/NFC, Applied Computer Vision

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Availability: flexible start in 2026 - 6-month final-year internship to be completed before Dec 2026 | Open to Europe & Asia |
Work Authorization: available upon request

PROFILE

- Final-year engineering student (ESILV/IFT) building end-to-end products from rapid prototyping and UX iteration to deployment.
 - Shipped an offline-first NFC-based PWA for field traceability (IndexedDB/Dexie, Web NFC) and maintained frequent releases (Vercel, CI/CD).
 - R&D foundation through computer-vision pipelines (Python/OpenCV) and ML experimentation (pose estimation + classification).
- Portfolio (WebGL 3D, FR): aure3479.github.io/Portfolio/ for a lighter view, see pinned projects on GitHub

WORK EXPERIENCE

PWA / NFC Developer Intern - Pass Abeilles (Beekeeping IoT Startup), Haute-Vienne, France Summer 2025

- Designed and shipped an offline-first PWA enabling NFC-based hive and supers traceability in the field.
- Built local-first persistence with IndexedDB (Dexie.js) to support operations with no connectivity and minimize data loss risk.
- Iterated UX through user research and rapid prototyping; reached 92% task completion rate during user acceptance testing (UAT).

Tech: React, TypeScript, Dexie.js, Web NFC API, PWA, Vercel

Software Developer Intern - Virobotic (Medical/Biotech Startup), Amiens, France Summer 2022

- Developed and optimized computer-vision pipelines to process large-scale medical device databases.
- Implemented parameter-detection algorithms for surgical instrument characteristics; improved detection accuracy by 18%.
- Performed statistical trend analysis on 1M+ records to refine search parameters and support product validation.

Tech: Python, OpenCV, Pandas, SQL

PROJECTS & LEADERSHIP

Creator / Lead Developer - TimbreForge (Voice-to-Haptics) (Open-source project) (ESILV project) 2025-2026

- Built a real-time voice-to-haptics pipeline mapping audio features to 6 vibrotactile channels for expressive feedback.
- Implemented BLE control of an ESP32 hub driving multiple LRA actuators and a desktop GUI for calibration and live control (replicable setup).

Tech: Python (real-time audio), BLE, ESP32, LRA haptics, desktop GUI

Lead Developer - Movement Profiling via Pose Estimation (ActionTypes-inspired) (ESILV project) 2024-2025

- Architected a modular pipeline (MediaPipe + OpenCV) for real-time capture, feature extraction, and classification.
- Reached 85% classification accuracy in current iteration.

Tech: Python, OpenCV, MediaPipe, PyTorch, scikit-learn, Flask, React

Product Builder (Kickstarter) - Tic-Tac Cheers (leather wine-glass markers) (Micro-business) — End-to-end: design → sourcing/manufacturing → sales; €500 revenue (110% target).

SKILLS

Core Software: JavaScript/TypeScript, React/Next.js, Flask, PWA, Web NFC API, IndexedDB (Dexie.js), CI/CD, Git/GitHub, Vercel

Data & Analytics: Python (Pandas, NumPy), SQL, statistical analysis, dashboards/reporting

Hardware: Arduino/ESP32, BLE, haptics (LRA actuators), sensor integration, KiCad (PCB), SolidWorks (CAD)

Tools: MS Office (Excel/PowerPoint), Figma, Unity, VS Code

EDUCATION

Pôle Universitaire Léonard de Vinci (ESILV), Paris/La Défense, France

Engineering Degree (Master's level), General Engineering (Major: Creative Technologies) - 2021-2026 (Expected)

Master of Science, Future Technologies (IFT) - 2024-2026 (Expected)

Design Thinking Certificate – 2023

LANGUAGES & INTERESTS

Languages: French (Native), English (C1), German (B1)

Interests: R&D innovation, robotics/IoT, human-centered AI, rapid prototyping, building products with measurable impact